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ORIGINAL PAPER

Dual-timescale hierarchical MADDPG for Multi-UAV cooperative search

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Abstract

Cooperative exploration conducted by multiple unmanned aerial vehicles (UAVs) facilitates parallelized reconnaissance over expansive territories, thereby optimizing the efficiency of target localization. This study investigates the challenge of coordinated search for sparsely located, initially undiscovered stationary targets by a fleet of UAVs constrained by limited perceptual capabilities. Effective resolution of this issue is pivotal for attaining rapid situational awareness in expansive, time-sensitive missions such as disaster mitigation and strategic intelligence gathering. Nonetheless, prevailing methodologies for multi-UAV search frequently encounter limitations in concurrently achieving exhaustive spatial coverage and elevated target acquisition efficacy. To overcome these deficiencies, this study introduces a dual-timescale hierarchical reinforcement learning paradigm tailored for collaborative multi-UAV search missions. The proposed Dual-Timescale Hierarchical Multi-Agent Deep Deterministic Policy Gradient (DTH-MADDPG) architecture incorporates a high-level strategic controller and an array of low-level decentralized agents, thereby enabling temporally stratified policy optimization. This framework facilitates a more nuanced equilibrium between macro-scale environmental coverage and micro-scale target identification than monolithic architectures. Empirical evaluations within simulated operational environments reveal that DTH-MADDPG markedly surpasses contemporary benchmark algorithms, demonstrating superior scalability, accelerated convergence rates, and heightened resilience.

Keywords Multi-agent reinforcement learning · Markov decision process · Multi-UAV cooperative search

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1 Introduction

Multi-UAV cooperative path planning (Xu et al. 2020) constitutes a complex and practically consequential problem in domains such as surveillance (Chen et al. 2022), search-and-rescue (Yanmaz 2023), and military applications (Liu et al. 2023). Conventional single-agent planning algorithms frequently prove inadequate in multi-agent contexts, primarily due to the combinatorial explosion of the joint state-action space and the imperative for inter-agent coordination.

Existing approaches to multi-UAV path planning under uncertainty manifest several critical limitations. Traditional graph-based and heuristic algorithms (Kelner et al. 2024; Qu and Fan 2024) efficiently generate shortest-path solutions in fully observable and static environments; however, they inherently lack inter-agent coordination mechanisms and typically necessitate complete re-planning in response to environmental perturbations. A considerable volume of research has explored swarm intelligence paradigms, with artificial potential field techniques and rule-based flocking models demonstrating empirical efficacy in decentralized

obstacle avoidance and formation cohesion (Zhang and Ali 2020; Perez-Carabaza et al. 2018). Despite their algorithmic simplicity and scalability, such heuristic swarm methods are susceptible to local minima entrapment and generally fail to optimize system-wide performance objectives.

An alternative research trajectory leverages game-theoretic frameworks. Potential game formulations conceptualize each UAV as a rational agent, with carefully designed utility functions ensuring that any Nash equilibrium aligns with a conflict-free trajectory configuration (Zhang et al. 2021). While these methods ensure distributed convergence to equilibrium states, such solutions may be globally suboptimal in terms of cumulative path cost or energy expenditure, and constructing appropriate payoff structures for complex, dynamic tasks remains nontrivial. Differential game models and auction-based mechanisms have also been explored for multi-agent path planning (Hou et al. 2023), providing principled strategies for conflict mitigation and task distribution. However, these formulations frequently presuppose complete system knowledge or static information, thereby limiting their applicability in non-stationary environments lacking prior models or requiring online adaptability. Metaheuristic and evolutionary algorithms present a distinct paradigm by conducting iterative, stochastic search for near-optimal solutions. Techniques such as genetic algorithms and particle swarm optimization have been successfully employed in multi-UAV routing scenarios (Ni et al. 2020; Hou et al. 2021), delivering feasible, collision-free paths while accommodating multiple optimization criteria, including route efficiency, fuel economy, and risk minimization. Despite their robustness in navigating complex and multi-modal search landscapes, these approaches typically involve considerable computational demands and are often relegated to centralized or offline planning regimes, posing challenges for real-time deployment and dynamic reconfiguration.

Recently, deep reinforcement learning (DRL) has demonstrated promise in addressing sequential decision-making under uncertainty for multi-agent systems (Lee and Frideikos 2022; Foerster et al. 2017; Wang et al. 2025, 2024). DRL-based path planning enables UAVs to learn navigation policies through trial-and-error interactions with a simulated environment, effectively capturing uncertainties and complex dynamics. For example, single-UAV DRL methods have shown the ability to avoid dynamic obstacles and unknown threats in real time (Foerster et al. 2017). An experience-memory Q-learning approach further improved single-UAV navigation in unknown terrain by leveraging prior exploration data (Du et al. 2019). Early attempts at applying RL to UAV path planning employed discretized state representations and shaped rewards (Luo et al. 2022), demonstrating feasibility but with limited scalability and lacking inter-agent coordination. Multi-agent RL frameworks, including centralized training with decentralized execution, have been proposed to address coordination by training cooperative policies or shaping joint reward functions (Li et al. 2021; Lee and Frideikos 2022). However, single-level RL approaches suffer from the curse of dimensionality in multi-UAV missions, resulting in slow convergence and suboptimal performance as the number of agents or planning horizon increases (Wang et al. 2021). Without additional structure, a monolithic policy struggles to simultaneously address high-level team coordination and low-level motion control. This limitation motivates a hierarchical reinforcement learning approach.

In this study, we propose an innovative hierarchical reinforcement learning framework designed for cooperative multi-UAV path planning. Diverging from conventional flat reinforcement learning paradigms and broadly scoped multi-agent hierarchical models, the proposed approach distinctly segregates the planning task into high-level spatial zone allocation and low-level motion planning. This stratified architecture, supported by a dual-tier reward mechanism, ensures that both strategic regional decisions and tactical navigation behaviors are optimized via harmonized yet level-specific feedback signals. By modularizing the decision-making process across complementary layers, the framework significantly advances scalability and inter-agent coordination: each policy concentrates on a constrained subproblem, thereby reducing overall complexity and minimizing policy conflicts. The principal contributions of this work are summarized as follows:

1. We propose a novel hierarchical reinforcement learning framework for multi-UAV cooperative search that decouples the problem into a high-level strategic coordinator and low-level tactical executors. The coordinator employs an Actor–Critic network for dynamic region allocation using constrained Voronoi optimization with provable coverage-time bounds, while the executors leverage Gumbel–Softmax policies for differentiable discrete action selection and potential-based reward decomposition to address credit assignment.
2. We introduce a unified, dual-phase reward architecture that adaptively balances exploration and exploitation via phase-specific entropy reduction metrics and spatiotemporal penalty terms, ensuring both load balancing and smooth policy transitions.
3. The proposed framework is theoretically grounded, with formal convergence guarantees relative to baseline performance metrics, and demonstrates a 40.7% faster exploration speed compared to centralized MADDPG methods.

The remainder of this paper is organized as follows. Section 2 provides a critical review of recent developments in multi-UAV path planning and cooperative search, examining heuristic and swarm-based strategies, game-theoretic

models, evolutionary metaheuristics, and deep reinforcement learning techniques. Section 3 elaborates on the proposed Dual-Timescale Hierarchical MADDPG (DTH-MADDPG) framework, including the design of the potential-based global reward function, the hierarchical policy structures, and the dual-timescale optimization protocol. Section 4 presents a comprehensive empirical evaluation, encompassing large-scale simulations, comparative benchmarks, ablation studies, and qualitative analyses of generated trajectories, thereby demonstrating the proposed method's efficacy, resilience, and scalability. Finally, Section 5 synthesizes the key contributions, reflects on their theoretical and practical significance, and outlines prospective avenues for advancing large-scale multi-agent aerial coordination.

2 Related work

In recent years, multi-UAV trajectory optimization and collaborative target search have attracted substantial academic attention, owing to their pivotal role in augmenting the autonomy, scalability, and operational efficiency of search and rescue missions. Through a synthesis of recent scholarly contributions (Lyu et al. 2023; Rahman et al. 2025), the present study delineates four prominent methodological paradigms that characterize the current research landscape: (i) heuristic and swarm intelligence-based algorithms, (ii) game-theoretic and auction-based coordination frameworks, (iii) evolutionary and metaheuristic optimization techniques, and (iv) deep reinforcement learning-driven approaches.

2.1 Heuristic and swarm-based methods

Early approaches to multi-UAV path planning have frequently relied on classical heuristics and swarm-inspired strategies that are relatively straightforward to implement. Graph search algorithms, such as A* and D*-Lite, have been extended to multi-agent scenarios through sequential planning or priority schemes to circumvent conflicts (Kelner et al. 2024). Although these methods guarantee optimal or feasible paths in static grid maps (Qu and Fan 2024), they are not inherently designed for inter-agent coordination; a path that is optimal for one UAV may conflict with that of another. To address agent interactions, researchers have developed artificial potential field (APF) methodologies, wherein UAVs are modeled as particles influenced by artificial repulsive and attractive forces (Zhang and Ali 2020). In a multi-UAV APF framework, obstacles exert repulsive forces while objectives impose attractive forces on each UAV, supplemented by inter-UAV repulsions to maintain spatial separation. APF-based planners facilitate decentralized obstacle avoidance and exhibit real-time performance, rendering them attractive for swarm deployments. Concurrently, several studies have employed neural networks as function approximators for path planning (Chaves et al. 2014), endeavoring to learn mappings from sensor inputs to collision-free waypoints; although these learned heuristics yield rapid execution, they require extensive training data and may not generalize effectively to novel scenarios. Nonetheless, potential field methods (and other fixed-response heuristics) are susceptible to local minima (e.g., a UAV may become trapped equidistantly between multiple obstacles) and can display oscillatory behavior, particularly in dynamic environments.

Swarm behavior models inspired by biological systems (Alfeo et al. 2019, ?) have also been employed in UAV teams. Flocking algorithms (Elmokadem and Savkin 2021), for example, utilize simple local rules (cohesion, separation, and alignment) to engender emergent coordinated movement within UAV groups. These methods naturally facilitate distributed formation flight and target encirclement without incurring substantial computational or communication overhead (Perez-Carabaza et al. 2018). Purely behavior-based swarm models, which depend exclusively on local interaction rules, lack mechanisms for global objective optimization and frequently yield formations or dispersions that do not necessarily minimize travel distance or energy expenditure.

Similarly, swarm-based heuristic methodologies, exemplified by the motion-encoded genetic algorithm with multiple parents (MEGA-MPC), which restructures global search trajectories via multi-parent crossover (Alanezi et al. 2022), provide advantages in terms of implementation simplicity and algorithmic resilience. MEGA-MPC embodies a swarm paradigm, wherein multiple UAVs collaboratively orchestrate their motion planning through collective evolutionary operators to enhance global search efficacy. Nevertheless, these heuristic paradigms often exhibit constrained adaptability, owing to the absence of embedded learning mechanisms capable of accommodating dynamically shifting objectives or unforeseen environmental perturbations.

2.2 Potential game and game-theoretic approaches

Game-theoretic methods cast the multi-UAV path planning problem as a strategic game, wherein each UAV (player) selects a trajectory to optimize a payoff function contingent upon the actions of all players. A prominent subset of these methods is potential games, which are structured such that any enhancement in an individual player's payoff is concomitant with an improvement in a global potential function. For multi-UAV coordination, researchers have formulated utility functions—typically comprising negative path cost augmented by collision penalties—that recast path planning as a cooperative potential game (Zhang et al. 2021). Within the potential game framework, UAVs iteratively refine their routes via best-response dynamics, with theoretical convergence to a Nash equilibrium assured. This

distributed paradigm empowers each UAV to independently optimize its trajectory, ultimately yielding a conflict-free configuration. However, the equilibrium's quality is highly contingent upon the payoff structure; frequently, the Nash equilibrium is only locally optimal, and attaining a globally optimal solution necessitates meticulous tuning or supplementary coordination mechanisms (Hou et al. 2023). Moreover, game-theoretic planners often presuppose rational behavior and complete awareness of other agents' states—assumptions that may not hold under uncertainty or partial observability.

Other game-theoretic approaches, such as auction-based task allocation and market-driven coordination (Qamar et al. 2023), decentralize decision-making by enabling UAVs to bid for tasks based on cost metrics, thereby expediting target distribution and sequencing. Nevertheless, decoupling task assignment from path planning may result in suboptimal routes. Similarly, differential game and optimal control formulations—employed, for instance, in modeling pursuit–evasion scenarios (Hou et al. 2023)—provide valuable theoretical insights but are frequently computationally prohibitive for systems comprising more than a few agents.

2.3 Evolutionary and metaheuristic algorithms

Evolutionary algorithms and other metaheuristic optimization techniques have been extensively applied to multi-UAV path planning due to their capacity to navigate complex, non-convex optimization landscapes. Genetic algorithms (GAs) are a prevalent choice, in which candidate path solutions—encoded as sequences of waypoints or control points—evolve over successive generations (Ni et al. 2020; Hou et al. 2021). The selection, crossover, and mutation operators inherent in GAs facilitate the joint optimization of UAV routes, potentially addressing multiple objectives such as total travel distance, threat exposure, and fuel consumption. Researchers have demonstrated that GA-based planners can yield high-quality solutions for multiple UAVs operating in obstacle-dense 3D environments (Hou et al. 2021). Nonetheless, GAs typically necessitate numerous generations to achieve convergence, and their computational overhead may render them impractical for real-time applications.

Swarm intelligence algorithms, notably Particle Swarm Optimization (PSO) (Phung and Ha 2021) and Ant Colony Optimization (ACO) (Perez-Carabaza et al. 2018), have been applied to cooperative path planning. PSO models candidate solutions as particles influenced by both cognitive and social factors, thereby facilitating waypoint optimization for multi-UAV missions (Zhang and Ali 2020). Conversely, ACO, inspired by ant foraging behavior, constructs paths probabilistically via pheromone trails on environmental graphs, fostering solution diversity and convergence toward optimal routes (Perez-Carabaza et al. 2018). Despite their efficacy in exploring intricate solution spaces under static conditions, these methods remain highly sensitive to parameter tuning and are predominantly implemented as offline optimizers, thereby curtailing their applicability in real-time dynamic environments. Metaheuristic techniques can be hybridized with alternative strategies to augment efficiency. For example, simulated annealing may be employed to refine UAV route plans following the derivation of an initial solution, and game-theoretic PSO hybrids have been proposed to ensure that collision avoidance constraints are maintained while optimizing performance (Hou et al. 2023). These hybrid approaches endeavor to alleviate the limitations of pure metaheuristics by incorporating problem-specific insights or coordination protocols. Nonetheless, a fundamental drawback persists: metaheuristic algorithms do not leverage historical experience, as each optimization is initiated anew.

2.4 Deep reinforcement learning and hierarchical RL approaches

The success of deep reinforcement learning (Fei et al. 2024) in single-agent domains has catalyzed its application to multi-UAV coordination challenges. In recent years, several studies have employed deep Q-networks (DQN) (Yan et al. 2020; Tang et al. 2024), policy gradient methods (Sabzekar et al. 2024), and actor-critic frameworks (Chen et al. 2024) for UAV path planning and control. For instance, variants of deep Q-learning have been utilized to empower a UAV to navigate uncharted environments and circumvent dynamic obstacles (Foerster et al. 2017; Yue et al. 2019). In multi-UAV settings, a centralized-training, decentralized-execution paradigm is frequently adopted: during training, a joint reward signal or aggregated state information is employed to learn cooperative policies, while during execution, each UAV relies on its local observation and a learned policy to make decisions (Li et al. 2021; Lee and Friderikos 2022). Techniques such as parameter sharing and consensus updating have been investigated to ensure that UAVs acquire symmetric policies conducive to implicit coordination. Although these flat DRL approaches exhibit enhanced adaptability compared to traditional methods, they encounter scalability challenges. The joint action space expands exponentially with the number of UAVs, and training may become unstable due to credit assignment issues (i.e., determining which agent's action precipitated a given reward) (Wang et al. 2021). Approaches such as mean-field multi-agent RL and value decomposition networks have been proposed to mitigate these challenges by approximating inter-agent interactions or decomposing value functions; however, their efficacy in complex, high-dimensional UAV tasks remains constrained by the inherent problem complexity.

In the domain of UAV path planning, a number of recent investigations have employed hierarchical reinforcement

learning (HRL) paradigms. Many extant HRL methodologies for multi-agent systems either model each agent's hierarchy in isolation (Han et al. 2019; Huh and Mohapatra 2023) or adopt leader-follower architectures (Kao et al. 2022). A representative instance leverages the MAXQ value function decomposition to disaggregate the overarching mission into manageable sub-tasks—such as regional exploration and local maneuvering—thereby reducing the complexity of each sub-problem and enabling more efficient learning (Yan et al. 2020). Another prominent strategy integrates a two-tier deep Q-network, wherein the upper layer selects intermediate waypoints and the lower layer executes navigational policies to reach them (Yao et al. 2017). These hierarchical formulations have demonstrated faster convergence rates and enhanced scalability to expansive operational environments relative to their flat reinforcement learning counterparts.

While hierarchical reinforcement learning architectures such as MAXQ and FeUdal RL (Ma and Wu 2020) have achieved notable success in single-agent scenarios, their extension to fully cooperative multi-agent contexts—particularly in large-scale multi-UAV search operations—remains underexplored. This gap underscores the imperative for a dual-timescale hierarchical framework capable of addressing the coordination, scalability, and non-stationarity challenges endemic to multi-agent environments. In this paper, we introduce an integrated, end-to-end hierarchical reinforcement learning framework explicitly designed for cooperative multi-UAV systems. In contrast to conventional HRL models that depend on manually specified sub-tasks, our framework features a high-level coordinator that adaptively allocates exploration objectives to maximize the global team reward, while decentralized low-level controllers efficiently execute the assigned directives. The adoption of a dense, structured reward hierarchy not only accelerates convergence compared to traditional sparse-reward paradigms but also enables effective gradient-based optimization of discrete high-level policies—a capability explored in the domain of neural models with discrete latent representations (Zhang et al. 2021), yet rarely exploited in multi-agent hierarchical planning.

3 Dual-timescale hierarchical MADDPG

Figure 1 presents an overview of the proposed **D**ual-**T**imescale **H**ierarchical MADDPG (**DTH-MADDPG**) framework for multi-UAV cooperative search, depicting the interaction dynamics between the high-level strategic coordinator and the low-level decentralized actors in executing coordinated search operations under static-target uncertainty. Specifically, it integrates a slow-timescale Actor–Critic coor-

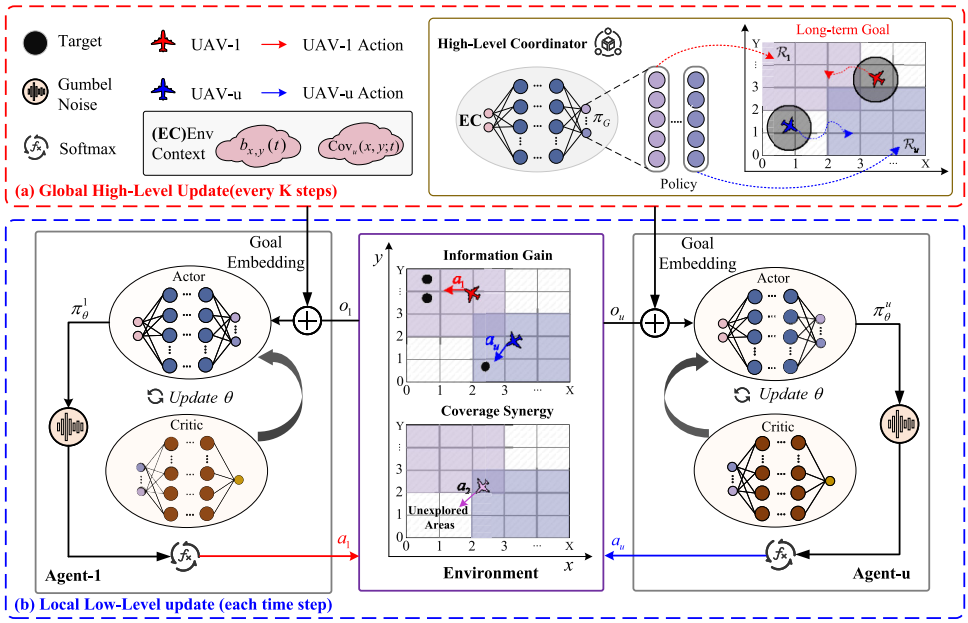

Fig. 1 Schematic of the Dual-Timescale Hierarchical MADDPG (DTH-MADDPG) for multi-UAV cooperative search under static-target uncertainty. **Top (red dashed):** The global coordinator receives $\{b_{x,y}(t)\}$ and $\{\mathrm{Cov}_u(x, y; t)\}$ as global environmental inputs, processes them through a CNN-based actor π_G, and outputs region assignment goal vectors g_u. **Bottom (blue dashed):** Each UAV augments its local observation o_t^u with g_u, inputs the concatenated vector $[o_t^u \oplus g_u]$ into a Gumbel–Softmax-regularized actor π_θ^u to generate logits and sample discrete heading actions $a_t^u \in \{-\alpha, 0, +\alpha\}$, evaluated by a centralized critic Q_ϕ^u. Policy and value functions are updated on a fast timescale at every step, while the high-level policy undergoes slower updates (every K steps), yielding coordinated, information-dense, and coverage-optimized trajectories

dinator that periodically partitions the operational domain and produces goal embeddings through a phase-blended information-coverage potential, while fast-timescale, Gumbel–Softmax–regularized actors exploit these embeddings to sample discrete heading commands under centralized critic supervision, thereby synthesizing global strategic planning with responsive, fine-grained local control.

3.1 Global reward for cooperative UAV search

In this subsection, we rigorously formulate the single-step global reward R_t for multi-agent reinforcement learning (MARL) within the framework of cooperative UAV search. Leveraging potential game theory, our formulation integrates information gain and spatial coverage into a unified reward function, thereby guiding UAVs in a decentralized decision-making paradigm toward near-optimal coordination for mitigating uncertainty in target localization.

Let the two-dimensional search area be discretised into the grid

$$\mathcal{D} = \{(x, y) \mid x = 1, \ldots, D_X,\ y = 1, \ldots, D_Y\}, \tag{1}$$

where $D_X \times D_Y$ is the total number of cells and each cell (x, y) may either contain a target or be empty. Time is treated as a discrete variable $t = 0, 1, 2, \ldots$.

For every cell (x, y) we maintain a posterior probability

$$b_{x,y}(t) = \mathbb{P}\big(\text{cell } (x, y) \text{ contains a target} \mid \mathcal{Z}_{1:t}\big), \tag{2}$$

where $\mathcal{Z}_{1:t}$ denotes the set of sensor observations collected from the start of the mission up to time t.

Under the static-target assumption, each cell posterior evolves solely through the UAV measurements:

$$b_{x,y}(t+1) = \text{BayesUpdate}\big(b_{x,y}(t),\ \mathbf{Z}_t\big), \tag{3}$$

where $\mathbf{Z}_t$ is the sensor data collected at time t and BayesUpdate $(\cdot, \cdot)$ fuses the prior $b_{x,y}(t)$ with the new measurements to obtain $b_{x,y}(t+1)$.

We consider N UAVs indexed by $\mathcal{U} = \{u_1, u_2, \ldots, u_N\}$. At each discrete instant t UAV u chooses a heading action $a_t^u \in \{-\alpha,\ 0,\ +\alpha\}$, where $\alpha > 0$ is the maximum allowable per-step yaw change (in degrees): $-\alpha$ denotes a left turn by α°, $+\alpha$ a right turn by α°, and 0 straight flight. The joint action vector is therefore

$$\mathbf{a}_t = \big(a_t^1,\ a_t^2,\ \ldots,\ a_t^N\big). \tag{4}$$

To determine whether grid cell $(x, y) \in \mathcal{C}$ is fully observed by UAV u at time t, we require the entire cell to lie inside the circular sensing footprint. Let $\partial\mathcal{C}(x, y)$ be the set of the four vertices of cell (x, y). The binary coverage indicator is therefore

$$\text{Cov}_u(x, y; t) = \begin{cases} 1, & \max\limits_{(x',y')\in\partial\mathcal{C}(x,y)} \big\|\mathbf{p}_u(t) - (x', y')\big\| \le R_{\text{sen}}, \\ 0, & \text{otherwise}, \end{cases} \tag{5}$$

where $\mathbf{p}_u(t) \in \mathbb{R}^2$ denotes the planar position of UAV u at time t and R_{sen} is its sensing radius.

To construct a unified single-step reward R_t for cooperative MARL, we introduce a potential function $\Phi(\mathbf{a}_t)$ that merges two elements: (i) an *information-gain* term measuring the reduction of occupancy uncertainty, and (ii) a *coverage potential* term penalising redundant overlap among UAV footprints. Denote these scalar quantities by $\Delta\mathcal{I}(\mathbf{a}_t)$ and $\text{CovPotential}\big(\mathbf{a}_t, b_{x,y}(t)\big)$, respectively; their combination yields the composite reward signal that drives coordinated multi-UAV behaviour.

The information-gain component is computed from an entropy metric. The *prior* entropy at time t is

$$H_{\text{prior}}(t) = \sum_{(x,y)\in\mathcal{C}} \Big[-b_{x,y}(t)\ \ln b_{x,y}(t) - \big(1-b_{x,y}(t)\big)\ \ln\big(1-b_{x,y}(t)\big)\Big], \tag{6}$$

where the sum runs over all grid cells $(x, y) \in \mathcal{C}$. Here $b_{x,y}(t)$ is the posterior occupancy probability defined earlier, and $\ln(\cdot)$ denotes the natural logarithm.

After the joint action $\mathbf{a}_t$ is executed and the new measurements $\mathbf{Z}_t$ are assimilated, the occupancy probabilities update to $b_{x,y}(t+1 \mid \mathbf{a}_t)$. The corresponding *posterior* entropy is

$$\begin{aligned} H_{\text{post}}(\mathbf{a}_t) = \sum_{(x,y)\in\mathcal{C}} \Big[&-b_{x,y}(t+1 \mid \mathbf{a}_t)\ \ln b_{x,y}(t+1 \mid \mathbf{a}_t) \\ &-\big(1-b_{x,y}(t+1 \mid \mathbf{a}_t)\big)\ \ln\big(1-b_{x,y}(t+1 \mid \mathbf{a}_t)\big)\Big], \end{aligned} \tag{7}$$

with the sum again taken over all cells $(x, y) \in \mathcal{C}$.

The resulting *information gain* is defined as

$$\Delta\mathcal{I}(\mathbf{a}_t) = H_{\text{prior}}(t)\ -\ H_{\text{post}}(\mathbf{a}_t), \tag{8}$$

which measures the reduction in overall occupancy uncertainty achieved by the joint action $\mathbf{a}_t$.

To encourage wide, non-redundant spatial sensing we define the *coverage potential*

$$\text{CovPotential}\big(\mathbf{a}_t, b_{x,y}(t)\big) = \sum_{(x,y)\in\mathcal{C}} \Big[\min\Big(1,\ \sum_{u=1}^{N} \text{Cov}_u(x, y; t)\Big)\, b_{x,y}(t)\Big], \tag{9}$$

where the inner factor $\min\left(1, \sum_{u=1}^{N} \mathrm{Cov}_u(x, y; t)\right)$ counts each cell at most once even when it is covered by several UAVs simultaneously, and the weight $b_{x,y}(t)$ biases the sum toward cells with a higher likelihood of containing a target, thus mitigating redundant observations.

3.2 Dual-timescale hierarchical MADDPG for Multi-UAV cooperative search

In this section, we introduce a hierarchical framework that segregates global region allocation from local action selection in multi-UAV cooperative search. Our architecture comprises two tiers: a high-level coordinator that employs an Actor–Critic (AC) network for strategic global decision-making, and a low-level executor that utilizes a Gumbel—Softmax based policy for discrete action selection. This decoupling facilitates comprehensive global planning while enabling agile local adaptation, thereby enhancing the overall efficiency of the search operation.

3.2.1 High-level region allocation

In the initial phase we partition the search domain $\mathcal{D} \subset \mathbb{R}^2$ into mutually disjoint regions so that every UAV receives a unique, non-overlapping area for immediate exploration. Let the planar positions of the N UAVs at time t be $\{\mathbf{p}_u(t)\}_{u=1}^{N}$. We adopt the standard Voronoi construction:

$$\mathcal{R}_u^0 = \left\{(x, y) \in \mathcal{D} \mid \|(x, y) - \mathbf{p}_u(t)\|_2 \le \|(x, y) - \mathbf{p}_v(t)\|_2,\ \forall\, v \ne u\right\}, \tag{10}$$

where $\mathbf{p}_v(t)$ is the position of UAV v ($v \in \{1, \ldots, N\},\ v \ne u$) and $\|\cdot\|_2$ denotes the Euclidean norm. By definition each region $\mathcal{R}_u^0$ comprises the points closest to UAVu, thereby minimising the travel time required to begin exploration.

Next, the regions are trimmed to reflect each UAV's actual sensing reach and flight budget. Assume every platform carries a sensor with range R_{sen} and moves at constant speed v. During the initial horizon T_1 a UAV can sweep, at most, the disc $\mathcal{B}\left(\mathbf{p}_u(t), R_{\mathrm{sen}} + vT_1\right)$, whose area is approximated by

$$A_u^{\max} = \pi\left(R_{\mathrm{sen}} + vT_1\right)^2. \tag{11}$$

The preliminary Voronoi regions $\{\mathcal{R}_u^0\}$ are therefore refined to $\{\mathcal{R}_u\}$ so that

$$\mathrm{Area}(\mathcal{R}_u) \le A_u^{\max} \quad \text{for all } u, \qquad \text{and} \qquad \bigcup_{u=1}^{N} \mathcal{R}_u = \mathcal{D}, \tag{12}$$

where $\mathrm{Area}(\cdot)$ denotes the planar area of a set.

The refinement is cast as the constrained optimisation problem

$$\min_{\{\mathcal{R}_u\}} \sum_{u=1}^{N} \left|\mathrm{Area}(\mathcal{R}_u) - A_u^{\max}\right|^2, \quad \text{s.t.} \quad \mathcal{R}_u \subset \mathcal{R}_u^0,\ ,\ \forall\, u \ne v. \tag{13}$$

To evaluate how thoroughly each refined region is explored, we define the coverage metric

$$\mathrm{RegionCov}(\mathcal{R}_u) = \frac{1}{|\mathcal{R}_u|} \sum_{(x,y)\in\mathcal{R}_u} \left(1 - \mathrm{Visited}(x, y; t)\right), \tag{14}$$

where $|\mathcal{R}_u|$ denotes the number of grid cells in $\mathcal{R}_u$ and

$$\mathrm{Visited}(x, y; t) = \begin{cases} 1, & \exists\, \tau \le t \text{ s.t. } \sum_{u=1}^{N} \mathrm{Cov}_u(x, y; \tau) \ge 1, \\ 0, & \text{otherwise.} \end{cases} \tag{15}$$

The binary flag $\mathrm{Visited}(x, y; t)$ therefore indicates whether cell (x, y) has been observed at least once by time t.

To accelerate the onset of exploration, we add a position-aware penalty. Let c_u be the geometric centre of region $\mathcal{R}_u$. Penalising the Euclidean offset $\|c_u - \mathbf{p}_u(t)\|_2$ discourages large deviations between the current UAV position and the centroid of its assigned region. The resulting initial-phase objective is

$$\{\mathcal{R}_u^{\mathrm{init}}\}_{u=1}^{N} = \arg\max_{\{\mathcal{R}_u\}} \sum_{u=1}^{N} \left[\mathrm{RegionCov}(\mathcal{R}_u) - \gamma\, \frac{\|c_u - \mathbf{p}_u(t)\|_2}{A_{\mathrm{env}}^{1/2}}\right], \tag{16}$$

where $\gamma > 0$ is a weighting coefficient and $A_{\mathrm{env}} = \mathrm{Area}(\mathcal{D})$.

Finally, if every UAV can sweep at most $A_u^{\max}$ within the horizon T_1, a sufficient condition for complete exploration is

$$\sum_{u=1}^{N} A_u^{\max} \ \ge\ A_{\mathrm{env}}, \tag{17}$$

subject to the individual constraints $\mathrm{Area}(\mathcal{R}_u) \le A_u^{\max}$.

To enable dynamic, long-horizon region allocation we embed an Actor–Critic (AC) architecture in the high-level planner. The global state—comprising the occupancy map $\{b_{x,y}(t)\}$ and the UAV positions $\{\mathbf{p}_u(t)\}$—is first processed by a deep CNN to capture spatial dependencies. The Actor produces, for each UAV u, a region label set $\mathcal{R}_u$ that is transformed into a binary indicator map $\mathbf{m}(\mathcal{R}_u) \in \{0, 1\}^{D_X \times D_Y}$, with $\mathbf{m}_{x,y}(\mathcal{R}_u) = 1$ iff $(x, y) \in \mathcal{R}_u$.

The indicator map is flattened to $\mathbf{m} \in \{0,1\}^{D_X D_Y}$ and passed through a two-layer multilayer perceptron

$$E_\psi : \{0,1\}^{D_X D_Y} \longrightarrow \mathbb{R}^{d_g}, \qquad E_\psi(\mathbf{m}) = \sigma_2\big(W_2\,\sigma_1(W_1\mathbf{m} + b_1) + b_2\big), \tag{18}$$

where $W_1 \in \mathbb{R}^{h \times D_X D_Y}$, $W_2 \in \mathbb{R}^{d_g \times h}$, $b_1 \in \mathbb{R}^h$, $b_2 \in \mathbb{R}^{d_g}$, $\sigma_1(\cdot) = \mathrm{ReLU}(\cdot)$, and $\sigma_2(\cdot) = \tanh(\cdot)$; h and d_g denote the hidden and embedding dimensions, respectively. The resulting fixed-length *goal embedding*

$$\mathbf{g}_u(t) = E_\psi\big(\mathbf{m}(\mathcal{R}_u)\big) \in \mathbb{R}^{d_g} \tag{19}$$

acts as a compact, differentiable directive for the low-level policy of UAVu.

Concurrently, the Critic evaluates a potential-based reward that reflects the allocation quality. In particular, the information gain for region $\mathcal{R}_u$ is

$$\mathcal{I}(\mathcal{R}_u) = H\big(b_{\mathcal{R}_u}(t)\big) - H\big(b_{\mathcal{R}_u}(t+1)\big), \tag{20}$$

with Shannon entropy

$$H\big(b_{\mathcal{R}_u}(t)\big) = -\sum_{(x,y)\in\mathcal{R}_u} \big[b_{x,y}(t)\ln b_{x,y}(t) + \big(1-b_{x,y}(t)\big)\ln\big(1-b_{x,y}(t)\big)\big]. \tag{21}$$

For long-term adaptation the high-level objective is reformulated to balance exploration and exploitation. In the refinement phase we maximise

$$\{\mathcal{R}_u^{\mathrm{ref}}\}_{u=1}^N = \arg\max_{\{\mathcal{R}_u\}} \sum_{u=1}^N \big[\mathcal{I}(\mathcal{R}_u) + \mathrm{Cov}(\mathcal{R}_u)\big] - \lambda\,\mathrm{Var}\big(\mathcal{I}(\mathcal{R}_1), \ldots, \mathcal{I}(\mathcal{R}_N)\big), \tag{22}$$

where $\mathrm{Cov}(\mathcal{R}_u)$ denotes the coverage potential of region $\mathcal{R}_u$ and $\lambda > 0$ penalises imbalances in information gain, thereby promoting load balancing across the fleet.

To smoothly transition from fast, breadth-first coverage to information-driven refinement, we introduce a **dynamically weighted reward**. A phase-transition factor $\beta(t) \in [0,1]$ is defined by

$$\beta(t) = 1 - \exp\big(-t/\tau_{\mathrm{init}}\big), \tag{23}$$

where the constant $\tau_{\mathrm{init}} = 200$ controls how quickly the weight shifts toward the refinement phase.

The resulting high-level reward at time t is

$$R_G^t = \underbrace{\big(1-\beta(t)\big)\,R_{\mathrm{init}}^t + \beta(t)\,R_{\mathrm{ref}}^t}_{\text{phase-blended term}} - \underbrace{\beta(t)\,\|\mathbf{g}_u^{t+1} - \mathbf{g}_u^t\|_2}_{\text{temporal smoothness}}, \tag{24}$$

where R_{init}^t and R_{ref}^t are defined in Eqs. (16) and (22), respectively. $\mathbf{g}_u^t$ denotes the goal embedding for UAV u at time t. The smoothness penalty limits abrupt changes in the goal embeddings, enhancing stability while the weighting factor $\beta(t)$ gradually shifts emphasis from rapid area coverage to information gain.

The Actor parameters θ_G are updated on a slower timescale (once every K environment steps). Using gradient descent we obtain

$$\theta_G^{k+1} = \theta_G^k - \eta_G\,\nabla_{\theta_G}\,\mathbb{E}\Big[\sum_{t=0}^{T/K} \gamma^{tK}\,R_G^t\Big], \tag{25}$$

where η_G is the learning rate, $\gamma \in (0,1)$ the discount factor, and the expectation is taken over trajectories generated by the current policy. Operating on this slow schedule allows the high-level coordinator to establish a rapid, balanced initial partition of the search domain and to refine that allocation iteratively, focusing UAV effort on regions with higher target probability.

Objective (16) rewards only previously unseen cells, while refinement term (22) penalises entropy–gain imbalance, pushing agents apart. The slow K-step update with a smoothness penalty stabilises partitions, thereby robustly reducing wasted flight time in both simulated and real-time deployments.

3.2.2 Low-level action selection

Given the high-level region allocation $\{\mathcal{R}_u\}_{u=1}^N$ and the corresponding goal embeddings $\{\mathbf{g}_u\}_{u=1}^N$ derived from the high-level coordinator, each UAV employs a decentralized policy for local motion planning within its designated region. To facilitate gradient-based optimization in a discrete action space, we recast the low-level decision-making process as a Gumbel–Softmax-regularized extension of the MADDPG framework.

In our formulation, the local policy for UAV u is conditioned on its assigned regional goal $\mathbf{g}_u$. At time t, the augmented state is defined as:

$$\tilde{\mathbf{s}}_t^u = \big[\mathbf{s}_t^u \oplus \mathbf{g}_u\big] \in \mathbb{R}^{d_s + d_g}, \tag{26}$$

where $\mathbf{s}_t^u = (\mathbf{p}_u(t), \mathbf{v}_u(t), \{\mathrm{Cov}_u(x,y;t)\})$ represents the local observation, comprising the UAV's position, velocity, and local coverage information, and $\oplus$ denotes vector concatenation. This goal-conditioned state enables the policy to adapt its exploratory behavior in accordance with the spatial characteristics of region $\mathcal{R}_u$.

The low-level actor of UAV u is a neural network $\boldsymbol{\pi}_\theta^u : \mathbb{R}^{d_s+d_g} \to \mathbb{R}^3$, parameterised by θ. Given the augmented state $\tilde{\mathbf{s}}_t^u$ it outputs a three-component logit vector

$$\mathbf{z}_t^u = \boldsymbol{\pi}_\theta^u(\tilde{\mathbf{s}}_t^u) = \big[z_-^u,\ z_0^u,\ z_+^u\big], \tag{27}$$

where z_-^u, z_0^u, and z_+^u correspond to the candidate turning actions $a_t^u \in \{-\alpha,\ 0,\ +\alpha\}$.

To obtain differentiable samples from the discrete action set we apply the Gumbel–Softmax trick. For each component $k \in \{-1, 0, +1\}$ an independent noise term $\gamma_k^u \sim \text{Gumbel}(0, 1)$ is drawn and added to the corresponding logit:

$$\tilde{z}_k^u = z_k^u + \gamma_k^u. \tag{28}$$

A temperature-controlled softmax is then applied,

$$g_k^u = \frac{\exp\left(\tilde{z}_k^u/\tau\right)}{\sum\limits_{k'\in\{-1,0,+1\}} \exp\left(\tilde{z}_{k'}^u/\tau\right)}, \qquad \tau > 0, \tag{29}$$

where the annealing parameter τ regulates the sharpness of the distribution. During the forward pass the discrete action is selected as

$$a_t^u = \arg\max_{k\in\{-1,0,+1\}} g_k^u, \tag{30}$$

while backward propagation employs the straight-through Gumbel–Softmax relaxation, ensuring that gradient information flows through the stochastic sampling step.

A centralised critic Q_ϕ^u evaluates the quality of the joint action vector $\mathbf{a}_t = (a_t^1, \ldots, a_t^N)$ given the global state. The latter comprises the collection of augmented local states $\{\tilde{\mathbf{s}}_t^u\}_{u=1}^N$ and the current occupancy map $b_{x,y}(t)$, $\mathbf{S}_t = \left(\{\tilde{\mathbf{s}}_t^u\},\ b_{x,y}(t)\right)$. The critic network is defined as

$$Q_\phi^u(\mathbf{S}_t, \mathbf{a}_t) = f_\phi\Big(\bigoplus_{u=1}^{N} \psi\big(\tilde{\mathbf{s}}_t^u\big) \ \oplus\ \text{CNN}\big(b_{x,y}(t)\big) \ \oplus\ \mathbf{a}_t\Big), \tag{31}$$

where $f_\phi(\cdot)$ is a multilayer perceptron parameterised by ϕ; $\psi(\cdot)$ is a shared feature extractor acting on each augmented state; and $\text{CNN}(\cdot)$ encodes the occupancy map into a spatial feature vector. This structure enables the critic to capture both inter-UAV dependencies and global search progress.

Each decentralised actor is trained by maximising the expected return via the policy-gradient rule

$$\nabla_{\theta^u} J(\theta^u) = \mathbb{E}_{\mathcal{D}}\Big[\nabla_{\theta^u} \log \boldsymbol{\pi}_\theta^u(a_t^u \mid \tilde{\mathbf{s}}_t^u) \ \cdot\ \nabla_{a_t^u} Q_\phi^u(\mathbf{S}_t, \mathbf{a}_t)\ \Big|\ a_t^u = \boldsymbol{\pi}_\theta^u(\tilde{\mathbf{s}}_t^u)\Big], \tag{32}$$

where the expectation is over minibatches drawn from the replay buffer $\mathcal{D}$. The Gumbel–Softmax reparameterisation ensures that gradients can flow through the discrete action selection step.

The critic parameters are updated via temporal-difference (TD) learning by minimising the squared Bellman error

$$\mathcal{L}(\phi) = \mathbb{E}_{(\mathbf{S}_t, \mathbf{a}_t, R_{t+1}, \mathbf{S}_{t+1})\sim\mathcal{D}}\Big[\big(y_{\text{target}} - Q_\phi^u(\mathbf{S}_t, \mathbf{a}_t)\big)^2\Big], \tag{33}$$

where the TD target is

$$y_{\text{target}} = R_{t+1} + \gamma\ Q_{\phi^-}^u\Big(\mathbf{S}_{t+1}, \{\boldsymbol{\pi}_{\theta^-}^v(\tilde{\mathbf{s}}_{t+1}^v)\}_{v=1}^N\Big). \tag{34}$$

Here $\gamma \in (0, 1)$ is the discount factor, while ϕ^- and θ^- are the slowly updated target-network parameters obtained via Polyak averaging.

To mitigate the multi-agent credit-assignment problem we decompose the global reward R_{t+1} into individual difference rewards. For UAV u the per-step reward is

$$r_{t+1}^u = \Phi(\mathbf{a}_t) - \Phi\big(\mathbf{a}_t^{-u}, a_{\text{null}}\big), \tag{35}$$

where $\Phi(\mathbf{a}_t) = \mu\ \Delta\mathcal{I}(\mathbf{a}_t) + (1-\mu)\ \text{CovPotential}(\mathbf{a}_t)$ is the potential function defined in Section 3.1; $\mathbf{a}_t^{-u}$ denotes the joint action with UAV u's component replaced by a null action a_{null}. This difference-reward formulation lets each agent estimate its own marginal contribution to the global objective, improving learning efficiency and stability.

Algorithm 1 Hierarchical enhanced MADDPG for Multi-UAV cooperative search.

Initialize: High-level AC network π_G, goal encoder E_ψ, low-level policies $\{\pi_L^u\}$ and critics $\{Q_\phi^u\}$, and replay buffer $\mathcal{D}$.
Initialize Environment: Set occupancy maps $\{b_{x,y}(0)\}$ and UAV positions $\{\mathbf{p}_u(0)\}$.
for each time step **do**
  for $k = 0$ to $T/K - 1$ **do** ▷ High-Level Planning (every K steps)
    Form global state from $\{b_{x,y}(kK)\}$ and $\{\mathbf{p}_u(kK)\}$.
    Compute region allocations $\{\mathcal{R}_u\} \leftarrow \pi_G(\cdot)$ and obtain goal embeddings $\mathbf{g}_u \leftarrow E_\psi(\mathcal{R}_u)$.
  end for
  for $t = 0$ to $T - 1$ **do** ▷ Low-Level Execution (each time step)
    for each UAV u **do**
      Form augmented state $\tilde{\mathbf{s}}_t^u = [\mathbf{s}_t^u \oplus \mathbf{g}_u]$.
      Compute logits $\mathbf{z}_t^u = \pi_L^u(\tilde{\mathbf{s}}_t^u)$.
      Sample discrete action a_t^u via Gumbel-Softmax.
    end for
    Execute joint action $\mathbf{a}_t$; observe next state and update $b_{x,y}(t+1)$.
    Compute individual rewards r_{t+1}^u using a potential-based difference function.
    Store transition $(\mathbf{S}_t, \mathbf{a}_t, \{r_{t+1}^u\}, \mathbf{S}_{t+1})$ in $\mathcal{D}$.
  end for
  High-Level Update: Update θ_G using aggregated high-level rewards.
  Low-Level Update: Sample mini-batch from $\mathcal{D}$ and update $\{\theta^u, \phi^u\}$ via TD error and policy gradient.
  Soft-update all target networks.
end for
Output: Final hierarchical policy $\{\pi_G, \pi_L^u\}_{u=1}^N$ for cooperative search.

Algorithm 1 encapsulates the proposed hierarchical learning framework, which operates across two distinct timescales. At the high level, region allocation is updated periodically—every K steps—during which the goal encoder E_ψ refreshes the embeddings $\{\mathbf{g}_u\}$ based on the latest occupancy maps

and UAV positions, and the high-level critic evaluates the long-term value of these allocations. At the low level, UAVs execute local actions at each time step using the current goal embeddings, and the resulting experience tuples $(\mathbf{S}_{kK+\tau}, \mathbf{a}_{kK+\tau}, R_{kK+\tau+1}, \mathbf{S}_{kK+\tau+1})$ for $\tau = 0, \ldots, K-1$ are stored in the replay buffer $\mathcal{D}$ for off-policy updates.

This hierarchical framework effectively decouples the global task of region allocation from local discrete action selection. The high-level Actor–Critic network ensures robust long-term planning and equitable region distribution, while the low-level Gumbel–Softmax-based policy enables precise, differentiable discrete action selection within each designated region.

4 Experiments

4.1 Parameter setting

In the simulation environment, we construct a grid map of dimensions 50×50, where each cell is initially assigned a uniform prior probability of 0.5 for containing a target. Four UAVs are randomly deployed on the grid at the outset of each trial. To evaluate the scalability of our proposed algorithm, the number of targets is varied across several distinct values (e.g., 10, 20), while the true target positions remain unknown to the UAVs throughout the training phase.

During training, each UAV executes actions for 3500 time steps. Following this phase, the trained policy is fixed, and the UAVs are allowed to explore new environments under identical parameter conditions. All experimental results are reported as averages over 100 independent runs to mitigate the effects of random initialization. Table 1 provides a comprehensive list of the hyperparameters and sensor specifications utilized in our experiments.

4.2 Performance benchmark

To assess the performance of the proposed DTH-MADDPG framework, we conduct comparative evaluations against a curated set of baseline algorithms encompassing value-based, heuristic, and policy-gradient methodologies. MADDPG is adopted as the primary learning-based benchmark due to its structural congruence with our approach—utilizing centralized training with decentralized execution, an actor—critic architecture, and experience replay—thereby enabling a controlled examination of the contributions from our hierarchical control schema and difference-based reward formulation. Furthermore, DQN and ACO are incorporated to represent non-hierarchical paradigms from deep reinforcement learning and swarm-based heuristic domains, respectively, alongside ME-RL, a recent entropy-regularized variant of MADDPG.

1. **DQN-Based Method** Harikumar et al. (2018): In this approach, each UAV independently deploys a Deep Q-Network (DQN) to estimate the expected return associated with each discrete action. Although the local observation and reward structures are analogous to those in alternative methods, the DQN relies solely on local state information for decision-making. Consequently, inter-UAV information sharing is minimal, potentially

Table 1 Key hyperparameters for the hierarchical MARL approach

Parameter	Symbol	Value
High-level Coordinator		
High-level strategy learning rate	η_G	1×10^{-3}
High-level strategy update interval	K	30
Initial stage time constant	τ_{init}	200
Region center position penalty weight	γ	0.3
Information gain variance penalty weight	λ	0.5
Low-level Executor		
Actor network learning rate	$\eta_{L,\text{actor}}$	1×10^{-4}
Critic network learning rate	$\eta_{L,\text{critic}}$	1×10^{-3}
Gumbel–Softmax initial temperature	τ_0	0.1
Temperature annealing rate	α_τ	0.995
Reward Function		
Information gain–coverage weight coefficient	μ	0.35
Global reward discount factor	γ_{global}	0.99
High-level reward smoothness penalty coefficient	β	0.3
Environment		
Effective sensor radius	R_{sen}	1.2
UAV movement speed	v	1.0

constraining scalability and cooperative efficacy when navigating larger or more intricate search environments.

2. **ACO-Based Method** Perez-Carabaza et al. (2018): Ant Colony Optimization (ACO) is utilized to steer the search process by modeling each UAV as an "ant" that deposits and follows pheromone trails. A three-dimensional pheromone matrix encapsulates the propensity for selecting each feasible heading across the grid for every UAV. While effective in smaller or more structured settings, ACO frequently encounters convergence issues and heightened computational overhead as the search space dimensionality and swarm size expand.
3. **MADDPG** Hou et al. (2023): This approach models the problem as a Markov Decision Process and leverages multi-agent reinforcement learning to enable resilient decision-making in uncertain and dynamic environments. Similar to our framework, it employs actor—critic updates and a replay buffer, thereby permitting a controlled evaluation of the hierarchical structure and the difference-reward mechanisms introduced in DTH-MADDPG.
4. **Maximum-Entropy RL(ME-RL)** Zhao et al. (2024): A communication-aware, energy-efficient extension of MADDPG that integrates spatial entropy, maximum-entropy exploration, and fuzzy logic to promote enhanced coordination and exploration efficiency.

4.3 Evaluation metrics

In addition to fundamental metrics such as coverage ratio and target detection count, we introduce convergence-based indicators to more precisely evaluate search efficiency. Specifically, to characterize the rate at which the UAV swarm attains operational effectiveness, we define the coverage convergence time T_{cove} and discovery convergence time T_{disc} as the earliest timesteps at which 80% of the environment is covered and 80% of the targets are located, respectively. These metrics offer a principled quantification of both spatial exploration and mission-critical responsiveness, directly correlating with system-level costs such as cumulative flight distance and energy expenditure.

Define the coverage convergence time T_{cove} as the earliest time step at which the spatial coverage ratio reaches 80%:

$$T_{\text{cove}} = \min\{t \in \mathbb{N} : C(t) \geq 0.8\}. \tag{36}$$

Let N_{target} denote the total number of targets and $N_{\text{found}}(t)$ the number detected by time t. We define the target discovery convergence time T_{disc} as the first time step when at least 80% of the targets have been detected:

$$T_{\text{disc}} = \min\{t \in \mathbb{N} : N_{\text{found}}(t) / N_{\text{target}} \geq 0.8\}. \tag{37}$$

4.4 Performance evaluation

In this section, we evaluate the scalability and convergence characteristics of the proposed DTH-MADDPG framework in comparison with two baseline methods: MADDPG and DQN. The experimental setup systematically increases the grid dimensions from 50×50 to 60×60 and 70×70, enabling a rigorous assessment of each algorithm's ability to manage increasingly expansive and complex state spaces. This analysis is critical for determining the extent to which the hierarchical structure of DTH-MADDPG maintains robust performance under escalating environmental complexity.

Figure 2 illustrates three subplots corresponding to the grid sizes: (a) 50×50, (b) 60×60, and (c) 70×70. The horizontal axis denotes the number of training time steps, while the vertical axis represents the average reward accumulated by the UAV swarm per step. Each trajectory indicates the mean performance over multiple independent trials, with shaded regions representing the standard deviation to capture inter-run variability.

Overall, the three subplots consistently demonstrate that DTH-MADDPG significantly outperforms both MADDPG and DQN across all tested map dimensions. During the early stages of training, the reward trajectories of DTH-MADDPG and MADDPG remain closely aligned, reflecting their shared Actor–Critic structure and the latent nature of DTH-MADDPG's hierarchical advantages. As training advances, however, DTH-MADDPG exhibits accelerated convergence, driven by the coordinated interplay between the high-level strategic policy and the low-level difference-reward shaping, which collectively promote more effective and targeted exploration behaviors.

Moreover, beyond achieving faster convergence, the shaded regions in each subplot reveal that DTH-MADDPG maintains consistently lower variance compared to the other two methods. This reduced variability arises from the hierarchical architecture's capacity to allocate explicit, goal-directed sub-tasks to individual UAVs, effectively curbing uncoordinated or adversarial behaviors and imparting greater stability to the collective learning dynamics.

During the later phases of training, DTH-MADDPG distinctly diverges from MADDPG, consistently sustaining a markedly higher average reward. The hierarchical framework empowers UAVs to dynamically decompose coverage tasks and reduce redundancy, an advantage that becomes increasingly pronounced with larger grid dimensions. This architectural design expedites the discovery of concealed targets and more efficiently mitigates global uncertainty relative to flat MARL baselines. While both methods exhibit comparable early-stage performance, the superior coordination facilitated by the hierarchical mechanism ultimately enables DTH-MADDPG to attain higher asymptotic rewards and more stable convergence behavior.

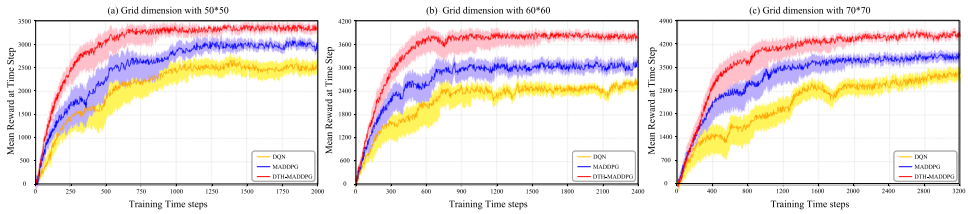

Fig. 2 Performance comparison of DQN, MADDPG, and DTH-MADDPG on grids of varying sizes: (a) 50 × 50, (b) 60 × 60, and (c) 70 × 70

Figure 3 displays three subplots corresponding to different grid sizes (50 × 50, 60 × 60, and 70 × 70). The x-axis denotes discrete time steps during which UAVs execute their respective heading actions, while the y-axis represents the coverage rate—defined as the proportion of grid cells that have been visited by any UAV.

The subplots clearly indicate that the ACO method persistently underperforms, hindered by its dependence on heuristic pheromone-based mechanisms that scale poorly in high-dimensional environments, resulting in slower and less stable coverage progression. By contrast, the deep reinforcement learning algorithms—DQN, MADDPG, ME-RL, and the proposed DTH-MADDPG—achieve substantially higher coverage rates within significantly fewer time steps, though their trajectories begin to diverge in the later stages. During the initial exploration phase (0–300 time steps), the DRL-based methods exhibit comparable behavior as agents engage in predominantly undirected exploration. As the episodes progress, DTH-MADDPG exhibits a significantly faster acceleration in coverage accumulation, attaining near-complete observation of the 50 × 50 environment by approximately 1500 time steps, whereas ME-RL, MADDPG and DQN typically require more than 2000 steps to reach a similar level of coverage.

These performance gains are attributable to the hierarchical control structure and the difference-based reward design employed in DTH-MADDPG. At the high level, global regions or subgoals are periodically reassigned through a potential-driven objective that balances spatial coverage with expected information gain, ensuring effective UAV dispersion and mitigating clustering behaviors. This facilitates a broader and more uniform observational footprint. At the low level, each UAV utilizes a Gumbel-softmax-based discrete actor guided by a locally shaped reward that quantifies its marginal contribution to global coverage. This difference reward, derived from the underlying potential function, promotes efficient local exploration and reduces trajectory redundancy. As a result, UAVs are better positioned to prioritize unexplored or high-uncertainty regions throughout the search process, thereby accelerating convergence and enhancing final coverage performance.

Figure 4 presents three subplots corresponding to grid sizes of 50 × 50, 60 × 60, and 70 × 70, respectively. The x-axis represents the number of time steps. During the early phase—particularly around 1000 time steps—DTH-MADDPG temporarily lags behind MADDPG, ME-RL, and DQN. This initial shortfall is attributable to the hierarchical search strategy: the high-level coordinator assigns macroscopic regions to UAVs with an emphasis on spatial balance and global exploration, which can lead to broader scanning patterns that defer immediate target detection. As time progresses, the interplay between structured region allocation and difference-based reward feedback allows UAVs to refine their low-level behaviors, increasingly focusing on high-uncertainty, unvisited areas. This facilitates a rapid acceleration in the target discovery rate, enabling DTH-MADDPG to outperform the baselines by achieving near-complete detection of all 20 targets well before 2000 time steps in the 50 × 50 environment. In contrast, MADDPG and DQN typically require in excess of 2000 steps to detect the majority of targets, reflecting weaker global coordination and more redundant exploration.

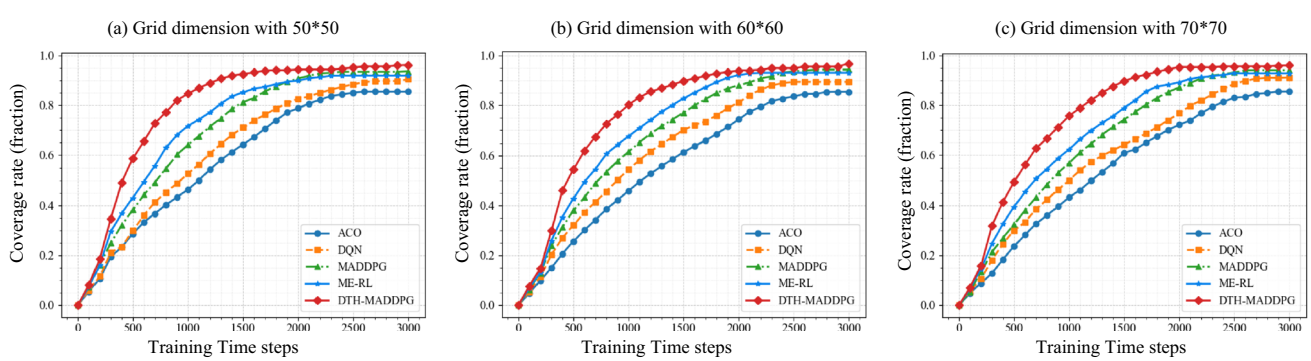

Fig. 3 Coverage rate comparison under varying grid sizes: (a) 50 × 50, (b) 60 × 60, and (c) 70 × 70

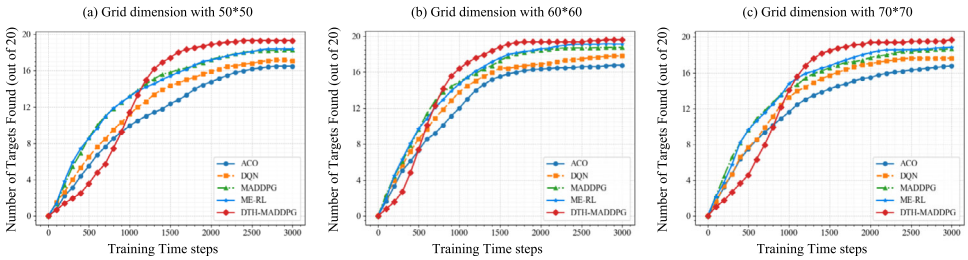

Fig. 4 Number of targets found by each method over time for three grid sizes: (a) 50 × 50, (b) 60 × 60, and (c) 70 × 70. The x-axis denotes the time steps, and the y-axis indicates the cumulative number of discovered targets out of 20 static targets randomly placed in the environment

Moreover, as the grid size increases from 50 × 50 to 70 × 70, the inherent complexity of the search space intensifies, leading to a general deceleration in target discovery across all evaluated methods. Despite this, DTH-MADDPG consistently retains a pronounced performance advantage, successfully identifying nearly all targets by approximately 1500 time steps even in the largest environment. This result underscores the efficacy of the hierarchical region allocation mechanism, which—when combined with potential-based reward shaping—facilitates efficient spatial decomposition and alleviates the scalability limitations that commonly hinder flat deep reinforcement learning approaches.

In the context of UAV search operations, a lower time step count signifies a more effective strategy, reflecting the algorithm's ability to rapidly traverse the environment while simultaneously locating the majority of targets. This assessment offers a direct measure of search strategy quality, wherein expedited spatial exploration and prompt target identification constitute key indicators of performance. Thus, two tables are provided to reflect performance under varying operational conditions. Table 2 reports results for scenarios containing 20 static targets, while Table 3 corresponds to configurations with 30 targets. In both tables, the columns labeled "50×50," "60×60," and "70×70" represent different grid dimensions, each subdivided into two metrics: coverage convergence time T_{cove} and the discovery convergence time T_{disc}. The downward arrow symbol (▼) adjacent to DTH-MADDPG entries emphasizes that lower values denote superior performance. Bold values indicate the best performance among the compared methods for each scenario.

The experimental findings demonstrate that the proposed DTH-MADDPG method consistently outperforms baseline approaches—including conventional MADDPG, DQN, and ACO—across a range of grid sizes and UAV configurations. Specifically, in the scenario involving 20 static targets and a team of 4 UAVs, averaged over 100 independent trials, DTH-MADDPG achieves an average speedup of 40% relative to MADDPG, 25% relative to MADDPG, 60% over DQN, and 240% compared to ACO. By integrating a high-level region assignment strategy with a low-level, difference-reward-

Table 2 coverage convergence time T_{cove} and the discovery convergence time T_{disc} for 20-target scenarios

UAV Num	Method	50*50		60*60		70*70		Total
		T_{cove}	T_{disc}	T_{cove}	T_{disc}	T_{cove}	T_{disc}	
UAV-4	**DTH-MADDPG**	**977▼**	**1176▼**	**1063▼**	**1319▼**	**1273▼**	**1507▼**	**7315▼**
	ME-RL	1423	1315	1617	1531	1616	1688	9190
	MADDPG	1647	1417	1773	1698	1896	1864	10295
	DQN	2015	1641	2236	1803	2456	2001	12152
	ACO	2572	2890	3278	3177	3984	3578	19479
UAV-6	**DTH-MADDPG**	**854▼**	**967▼**	**927▼**	**1131▼**	**1131▼**	**1296▼**	**6306▼**
	ME-RL	1287	1114	1432	1221	1465	1345	7864
	MADDPG	1415	1231	1578	1317	1699	1554	8794
	DQN	1814	1398	1900	1507	2096	2015	10730
	ACO	2036	1678	2202	2147	2309	2518	12890
UAV-8	**DTH-MADDPG**	**731▼**	**743▼**	**844▼**	**943▼**	**971▼**	**1097▼**	**5329▼**
	ME-RL	1113	1256	1232	1542	1324	1656	8123
	MADDPG	1277	1465	1317	1668	1434	1765	8926
	DQN	1633	2119	1713	1403	1837	1883	10588
	ACO	1770	1543	1834	2013	2130	2317	11607

Lower values (▼) indicate better performance. Columns summarize results across grid sizes (50×50, 60×60, 70×70), UAV team sizes (4, 6, 8), and four methods

Table 3 coverage convergence time T_{cove} and the discovery convergence time T_{disc} for 30-target scenarios

UAV Num	Method	50*50		60*60		70*70		Total
		T_{cove}	T_{disc}	T_{cove}	T_{disc}	T_{cove}	T_{disc}	
UAV-4	**DTH-MADDPG**	**1034▼**	**1197▼**	**998▼**	**1387▼**	**1273▼**	**1412▼**	**7301▼**
	ME-RL	1531	1339	1572	1599	1731	1856	9628
	MADDPG	1620	1480	1693	1732	1840	1956	10321
	DQN	2103	1596	2309	1739	2357	2016	12120
	ACO	2606	2793	3378	3155	3929	3659	19520
UAV-6	**DTH-MADDPG**	**896▼**	**930▼**	**982▼**	**1113▼**	**1191▼**	**1222▼**	**6334▼**
	ME-RL	1314	1253	1297	1256	1521	1416	8057
	MADDPG	1405	1328	1494	1366	1632	1578	8803
	DQN	1879	1352	1928	1448	2186	2003	10796
	ACO	1953	1733	2180	2210	2348	2501	12925
UAV-8	**DTH-MADDPG**	**752▼**	**687▼**	**934▼**	**938▼**	**1004▼**	**1021▼**	**5336▼**
	ME-RL	1216	1324	1232	1598	1276	1688	8334
	MADDPG	1345	1443	1332	1715	1395	1847	9077
	DQN	1562	2128	1625	1469	1784	1888	10456
	ACO	1746	1614	1743	2025	2158	2254	11540

guided exploration mechanism, DTH-MADDPG functions across dual timescales to achieve efficient coordination. This hierarchical architecture ensures distinct spatial responsibilities among UAVs, effectively reducing redundant coverage and accelerating convergence in both spatial exploration and target detection. Although traditional methods may yield similar performance in early training stages, the additional coordination layer in DTH-MADDPG enables substantially more rapid and sustained performance improvements. Importantly, as the operational environment scales in complexity, DTH-MADDPG maintains its advantage—often reaching 80% detection thresholds by approximately 1500 time steps—whereas alternative methods require significantly more time. These results underscore the framework's capacity to enhance individual UAV efficiency and optimize the overall effectiveness of cooperative search missions.

4.5 Ablation study

To further substantiate the contribution of principal components in our DTH-MADDPG framework, we conduct ablation experiments by selectively omitting each component and contrasting the resultant performance against that of the complete model. In particular, two crucial facets are identified: (i) the coverage reward term incorporated into the reward function, and (ii) the high-level regional allocation strategy facilitating coordinated UAV exploration. Examining the individual impact of these features provides deeper insights into their respective roles in expediting area coverage and target detection within complex search environments.

In the first ablation variant, we remove the coverage reward component from the reward function, retaining only the information gain term. This variant, denoted "DTH-MADDPG w/o Cov," evaluates the impact of the coverage potential in minimizing redundant sensing and encouraging broad exploration. In the second variant, we eliminate the high-level allocation strategy and revert to a flat structure where each UAV relies solely on its low-level policy ("DTH-MADDPG w/o HL"). This variant investigates how pivotal hierarchical coordination is for efficient area partitioning and reducing overlaps. We compare both ablated models with the full DTH-MADDPG under identical conditions. Table 4 presents the required time steps to reach 80% area coverage (Conv) and 80% target discovery (Target) with four UAVs in environments of size 50×50, 60×60, and 70×70. Lower time steps values indicate superior performance.

As shown in Table 4, the complete DTH-MADDPG method consistently surpasses both ablation variants. Excluding the coverage reward leads to a notable increase in the required time steps, underscoring the coverage potential's

Table 4 Ablation study of coverage convergence time T_{cove} and the discovery convergence time T_{disc} for a 4-UAV team on 50×50 and 70×70 grids

UAV Numbers	Method	50×50		70×70	
		T_{cove}	T_{disc}	T_{cove}	T_{disc}
UAV-4	**DTH-MADDPG (Full)**	**977▼**	**1176▼**	**1273▼**	**1507▼**
	DTH-MADDPG w/o Cov	1250	1317	1526	1765
	DTH-MADDPG w/o HL	1420	1611	1693	1847

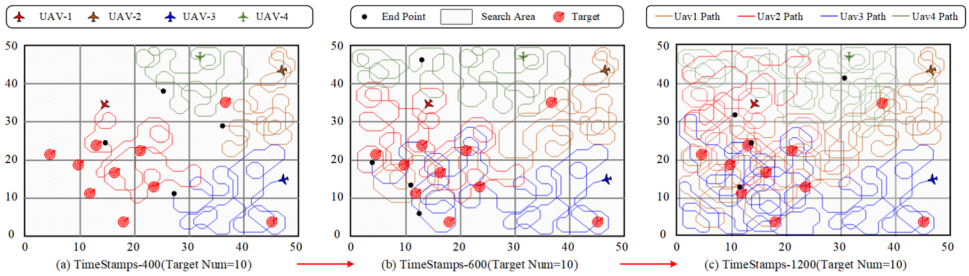

Fig. 5 Case study of 4 UAVs in a 50 × 50 grid at $t = 400, 800, 1200$. UAV paths are color-coded; red circles indicate targets

critical role in promoting efficient, non-redundant exploration. Likewise, discarding the high-level allocation strategy induces even greater performance degradation, emphasizing the importance of hierarchical region assignment for coordinated multi-UAV search.

4.6 Case study

This case study illustrates the real-time coordination capabilities of the DTH-MADDPG framework, highlighting both the global coverage strategy and the adaptive, context-sensitive nature of the underlying search policy. While earlier experiments focused on quantitative performance indicators such as coverage rate, target discovery, and cumulative reward, a qualitative analysis of UAV trajectories provides additional insight into how the hierarchical architecture and difference-based rewards influence agent behavior over time. Figure 5 presents three snapshots of the search progression within a 50 × 50 environment, captured at distinct time steps—(a) 400, (b) 800, and (c) 1200. The four UAVs are represented by color-coded trajectories (blue, orange, green, and red), and randomly distributed targets are denoted by red circles. "End Point" markers identify each UAV's position at the respective time step, while the "Search Area" outlines the operational boundaries. A comparative analysis of the subfigures reveals the UAVs' evolving spatial coverage patterns, characterized by a progressive expansion of explored regions, a reduction in trajectory overlap, and increased concentration of search activity in areas with elevated target likelihood.

In subfigure (a), at $t = 400$, the hierarchical policy has already partitioned the search space at a macro level, incentivizing each UAV to occupy distinct areas and thus minimize redundant coverage at an early stage. This high-level allocation allows each UAV to assume responsibility for a different sector of the grid, expediting the initial exploration. By $t = 800$ (subfigure (b)), more than 70% of the environment has been surveyed, as indicated by the nearly comprehensive network of color-coded flight paths traversing the grid. The difference-based reward continues to encourage UAVs to move into unvisited cells rather than clustering in previously explored sections.

Finally, at $t = 1200$ (subfigure (c)), the UAVs converge on regions deemed more likely to harbor remaining targets, exhibiting denser flight paths in those areas. This behavior results from the synergy of local Gumbel–Softmax exploration—where each UAV refines its own policy—and global hierarchical oversight, which directs overall coverage. As a consequence, high-probability cells receive adequate scrutiny without compromising the breadth of the search. Through this coordinated strategy, the swarm effectively balances comprehensive exploration with targeted investigation of unconfirmed cells, thereby maximizing the likelihood of swift target detection.

Beyond approximately 2000 time steps, the high-level allocator has effectively saturated global coverage; at this stage, the information-gain component becomes dominant, attracting multiple agents toward shared entropy peaks. In the absence of an explicit inter-agent repulsion mechanism—particularly when assigned regions contract to scales smaller than the UAVs' sensing footprint—this leads to increased trajectory overlap as agents converge on the same high-uncertainty areas.

5 Conclusion

In this work, we propose a hierarchical reinforcement learning framework for cooperative multi-UAV path planning. The framework employs a two-tier policy structure comprising a high-level policy that allocates spatial regions to individual UAVs and low-level policies that perform localized exploration within these regions, guided by structured reward shaping. This hierarchical decomposition enhances scalability, facilitates inter-agent coordination, and improves credit assignment relative to flat RL approaches. Experimental results on representative scenarios validate the efficacy of the proposed method, demonstrating superior exploration efficiency and coordination compared to baseline models. Future work will focus on integrating short-range repulsive

shaping and hierarchical value functions to maintain inter-UAV separation and reduce path overlap during the later stages of mission execution, particularly in environments with densely clustered targets.

Data Availability Data will be made available on request.

Declarations

Competing Interests The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Ethical Approval This study did not involve human participants or animals, and ethical approval was not required.

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